

# FitNourish.AI: A Wearable-Integrated AI Framework for Personalized Nutrition, Health Risk Prediction, and Travel-Aware Grocery Optimization

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**Abstract**—FitNourish.AI addresses fragmented digital nutrition support by integrating personalized nutrition prediction, health risk assessment, wearable data, and grocery optimization into one AI-driven framework. The system uses an XGBoost multi-output regression model to estimate calorie and macronutrient needs from demographic, physiological, activity, and disease-related features. An XGBoost classifier predicts health risk levels, while an MLP-based product matching module and a travel-aware optimization module recommends cost-effective grocery purchases from local retailers. Experimental results show strong performance, including an R-squared value of 0.957 for nutrition prediction, 96.60 percent accuracy for health risk classification, and 12 to 25 percent grocery cost reduction.

**Keywords**— *AI, XGBoost, Nutrition, Health Risk, Wearables, Grocery Optimization.*

## I. INTRODUCTION

Personalized dietary management has become increasingly important due to the growing impact of chronic non-communicable diseases such as diabetes, hypertension, and cardiovascular disorders. These conditions require dietary decisions that are not only nutritionally appropriate but also responsive to individual health status, lifestyle behavior, and daily activity patterns. Recent developments in mobile health, artificial intelligence, and machine learning have created new opportunities for delivering personalized and data-driven nutrition support through digital platforms [2]–[5]. However, many existing nutrition systems still depend heavily on static user inputs or generalized dietary equations, which may not fully reflect real-time physiological and activity variations in users' daily lives [6], [7].

Existing research has separately explored AI-based diet recommendations, mobile nutrition monitoring, health-risk prediction, and grocery-related decision support. Machine learning models have been used to estimate nutritional

requirements and support personalized diet planning [1], [5], while mobile health applications have improved accessibility to dietary self-monitoring and health guidance [2]. Similarly, disease prediction models have shown strong potential for identifying health risks using clinical and lifestyle-related attributes [4]. Despite these advances, most existing systems remain fragmented. Nutrition planning, health risk assessment, and grocery purchasing are usually treated as independent functions, creating a gap between computational health intelligence and practical daily food decisions.

This limitation is particularly important in real-world grocery purchasing. Even when a user receives accurate calories and macronutrient recommendations, they still need to decide which food products to buy, which supermarket to visit, and whether cheaper products justify additional travel. Prior grocery recommendation and e-grocery logistics studies have considered context-aware recommendations, product availability, delivery efficiency, and last-mile optimization [8], [9]. However, these systems are generally not connected to disease-aware nutrition targets or personalized health-risk predictions. As a result, users may receive cost-based or convenience-based grocery suggestions that do not fully align with their health needs.

Another major challenge is reliable cross-retailer product comparison. Supermarket product catalogues often differ in product names, brands, unit sizes, categories, and package descriptions. Therefore, comparing equivalent products across retailers requires product matching or entity resolution methods that can identify duplicate or similar records accurately [10], [11]. In addition, practical grocery recommendations should consider travel cost rather than selecting products only based on item price. Distance estimation methods such as the Haversine formula [12] and route-based purchasing optimization models such as the Traveling Purchaser Problem (TPP) [13] provide useful foundations for balancing product cost and travel effort.

However, their integration into a health-aware nutrition recommendation pipeline remains limited.

To address these gaps, this paper presents **FitNourish.AI**, a unified AI-driven framework that integrates personalized nutrition prediction, wearable-informed data acquisition, health risk stratification, and travel-aware grocery optimization. The proposed system collects demographic, physiological, disease-related, and activity-based user data through manual input and wearable synchronization. Baseline energy requirements are supported by established energy expenditure equations [5], [6] and refined using an XGBoost-based multi-output regression model to predict personalized calorie and macronutrient targets [12]. In parallel, an XGBoost classifier is used to categorize users into Low, Medium, and High health-risk levels based on key cardiometabolic indicators [4], [12]. The grocery optimization module applies machine learning-based product matching and travel-aware cost analysis to recommend practical grocery purchasing options across Sri Lankan retailers.

The main contributions of this work are fourfold. First, it proposes a closed-loop framework that connects user health data, nutrition prediction, health-risk assessment, and grocery decision support within a single system. Second, it introduces a multi-disease adaptive nutrition modeling approach that incorporates wearable-derived activity data and physiological features. Third, it integrates predictive health analytics to support proactive dietary decision-making. Fourth, it develops a Sri Lankan retail-focused grocery optimization module that combines product matching, availability-aware price comparison, and travel-cost-aware single-store or multi-store recommendations. By linking personalized health intelligence with real-world grocery purchasing, FitNourish.AI addresses a key limitation of existing isolated nutrition and grocery systems.

## II. LITERATURE REVIEW

Personalized nutrition and mobile health (m-Health) systems have become important research areas due to the increasing need for adaptive dietary support among users with different lifestyles and health conditions. Early nutrition recommendation systems commonly depend on rule-based calculations and standard energy requirement equations to estimate calorie and macronutrient needs. Although these methods provide useful baseline values, they are limited because they do not sufficiently respond to real-time changes in activity level, physiological status, disease condition, and user behavior. Recent studies have therefore explored artificial intelligence and machine learning techniques to improve personalization. Wang *et al.* [2] demonstrated the use of smartphone-based m-Health and machine learning for nutrition service delivery, while Amadeh *et al.* [1] presented intelligent diet recommendation as a promising direction for personalized dietary management. Zhuang *et al.* [5] further showed that machine learning can support user-specific nutrition recommendation, and omics-integrated approaches have extended personalization by considering biological and ethical dimensions [3]. However, these systems mainly focus on producing dietary recommendations and often do not connect nutrition prediction with health-risk forecasting or practical grocery-level decisions.

Health risk prediction is another closely related area in preventive healthcare. Machine learning classifiers have been used to identify disease risks from physiological, clinical, and lifestyle features. Patel *et al.* [4] showed that smartphone-based data can support disease prediction, highlighting the potential of predictive analytics for early risk identification. Such approaches are useful for categorizing users into different risk levels based on indicators such as blood pressure, glucose level, cholesterol, activity behavior, and other health-related attributes. Nevertheless, most health forecasting models operate as independent prediction modules. They may inform a user about potential risk, but they rarely translate that prediction into personalized nutrition guidance or purchasing decisions. This separation limits practical usefulness, especially for individuals managing chronic conditions such as diabetes and hypertension, where daily food selection should be aligned with both nutritional requirements and health-risk status.

A separate body of work has investigated grocery recommendations, price comparison, and e-grocery logistics. SmartGrocer, for example, explored context-aware grocery support by considering user preferences and product-related information [8]. Other studies on e-grocery logistics have examined store selection, delivery efficiency, travel distance, and last-mile optimization [9]. These studies show that grocery purchasing is influenced not only by item price but also by product availability, store location, and travel effort. However, most grocery recommendation systems are designed primarily around cost reduction, convenience, or logistics efficiency. They generally do not consider disease-aware nutrition targets, wearable-derived activity data, or predicted health-risk levels. Therefore, a recommendation may be economically attractive but still unsuitable for the user's personal health needs.

Reliable product matching is also essential when grocery recommendations involve multiple retailers. Product catalogues across supermarkets often differ in naming conventions, brand descriptions, unit sizes, package formats, and category labels. As a result, direct price comparison can be inaccurate unless equivalent products are correctly identified. Entity resolution and duplicate detection techniques provide a foundation for this task. Elmagarmid *et al.* [10] reviewed duplicate record detection methods, while Christen [11] presented detailed concepts and techniques for record linkage and data matching. These methods are highly relevant for cross-retailer grocery comparison because incorrect matching can mislead the optimization process and produce unreliable recommendations.

Travel-aware decision-making further strengthens the practicality of grocery optimization. Selecting the cheapest product alone is not always optimal if the travel cost is high. Distance estimation methods such as the Haversine formula [12] and purchasing optimization models such as the Traveling Purchaser Problem (TPP) [13] provide useful foundations for balancing product prices with travel effort. Recent work has also examined grocery access and travel behavior. Yu [14] showed that environmental awareness and walkability perception can influence walking behavior to

grocery stores, suggesting that accessibility and travel context should be considered in grocery-related systems. However, these travel-aware and sustainability-focused studies are usually disconnected from AI-based nutrition prediction and health-risk analysis.

Overall, existing research can be grouped into three mostly separate streams: personalized nutrition systems that generate dietary targets, health forecasting models that predict risk levels, and grocery optimization systems that improve purchasing efficiency. Each stream provides useful methods, but their separation creates a gap between computational health intelligence and real-world dietary action. To address this gap, FitNourish.AI integrates wearable-informed nutrition prediction, health risk classification, product matching, and travel-aware grocery optimization into one closed-loop framework. Unlike standalone nutrition systems, it considers activity and disease-related features. Unlike standalone risk prediction models, it connects risk output with dietary and grocery decisions. Unlike conventional grocery optimizers, it incorporates user-specific health requirements together with availability, price, and travel cost. This unified design makes the proposed framework more suitable for practical personalized nutrition support in the Sri Lankan retail context.

### III. METHODOLOGY

The proposed **FitNourish.AI** framework was developed as a closed-loop AI-driven decision-support system that connects user health data, personalized nutrition prediction, health-risk classification, and travel-aware grocery purchasing. Unlike conventional nutrition applications that mainly provide static calorie estimates, the proposed system uses wearable-informed activity data, machine learning models, and grocery optimization to generate practical recommendations that support both health and daily food purchasing decisions. To improve readability in the IEEE two-column format, the methodology is presented through focused functional modules instead of a large overall architecture diagram, which can become unreadable when compressed.

#### A. Data Acquisition and Preprocessing

User data were collected through a hybrid mechanism consisting of manual input and wearable-based synchronization. Manual inputs included age, gender, height, weight, dietary goal, and health conditions such as diabetes and hypertension. Wearable-informed inputs included step count, active minutes, heart rate, and active calories retrieved through Health Connect where permission was available. This approach reduces manual entry while allowing the system to reflect daily activity variation in nutrition prediction.

Grocery product and price data were collected from Sri Lankan supermarket sources, mainly Keells and Cargills Food City. A hybrid collection approach was used, combining API-based extraction where available and browser automation for dynamic retailer pages. The collected attributes included product name, brand, category, unit size, price, availability status, retailer, outlet details, and timestamp. These records were stored in a structured database to support the latest price retrieval and cross-retailer product comparison.

Preprocessing was applied separately to user, wearable, and grocery data. Numerical attributes were range-checked and normalized, while categorical attributes such as gender, dietary goal, and disease status were encoded using one-hot encoding. Grocery records were cleaned through lowercasing, punctuation removal, whitespace normalization, unit standardization, duplicate filtering, and category alignment. Missing wearable values were handled using available manual inputs or previous valid records where appropriate. These preprocessing steps improved data consistency, model stability, and the reliability of product matching.

Table I summarizes the main data layers used in FitNourish.AI and the preprocessing operations applied before model training and grocery optimization. User and wearable data were validated, encoded, and normalized, while grocery data were cleaned and standardized to support reliable product matching and travel-aware recommendations.

TABLE I. SUMMARY OF DATA SOURCES AND PREPROCESSING STEPS

| Data Layer               | Main Attributes                                                          | Preprocessing Applied                                    |
|--------------------------|--------------------------------------------------------------------------|----------------------------------------------------------|
| User profile             | Age, gender, height, weight, goal                                        | Validation, encoding, normalization                      |
| Health and wearable data | Diabetes, hypertension, steps, heart rate, active calories, stress score | Range checking, synchronization, missing-value handling  |
| Grocery catalogue        | Product name, brand, unit, price, availability                           | Text cleaning, unit standardization, duplicate filtering |
| Store and travel data    | Retailer, outlet, location, distance                                     | Distance estimation and travel-cost preparation          |

#### B. Baseline Nutritional Estimation

The nutrition module first estimates baseline energy requirements using accepted physiological equations. Basal Metabolic Rate (BMR) was calculated using the Mifflin–St Jeor equation [6]:

$$BMR = 10W + 6.25H - 5A + S$$

where W denotes body weight in kilograms, H denotes height in centimeters, A denotes age in years, and S represents the gender-specific constant. Total Daily Energy Expenditure (TDEE) was then estimated using an activity factor [7]:

$$TDEE = BMR \times AF$$

where AF represents the user's activity level. When wearable-derived active calories were available, they were used as an adjustment signal rather than being directly added without control. This prevents overestimation caused by double-counting activity expenditure. The calorie target was then adjusted according to the user's goal, such as weight loss, maintenance, or weight gain. Macronutrient targets were derived using standard energy conversion factors, where protein and carbohydrates provide 4 kcal/g and fat provides 9 kcal/g. The final nutrition output is represented as:

$$Y = [Calories, Protein_g, Carbohydrate_g, Fat_g]$$

This baseline layer provides interpretability and ensures that the machine learning-based output remains grounded in accepted nutritional estimation principles

### C. Personalized Nutrition Prediction

An XGBoost-based multi-output regression model was used to predict daily calorie requirement, protein intake, carbohydrate intake, and fat intake [16]. XGBoost was selected because it performs effectively on structured tabular datasets and can capture nonlinear relationships between demographic, physiological, disease-related, and activity-based features.

The input feature set included age, gender, height, weight, dietary goal, diabetes status, hypertension status, step count, active minutes, active calories, heart rate values, and engineered attributes such as Body Mass Index (BMI), activity ratio, and heart-efficiency indicators. The dataset was divided into training and testing subsets using an 80:20 split. Hyperparameter tuning and cross-validation were applied to improve generalization and reduce overfitting.

The model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination  $R^2$ . MAE was used to measure the average prediction error, RMSE was used to penalize larger deviations, and  $R^2$  was used to measure how well the model explained variation in the predicted nutrition outputs.

The nutrition module presents the predicted outputs through a user-readable interface. After user profile entry and wearable data synchronization via Health Connect, the system updates the user's health data and generates personalized calorie and macronutrient targets. Fig. 1 illustrates this workflow from input and synchronization to the final nutrition summary.

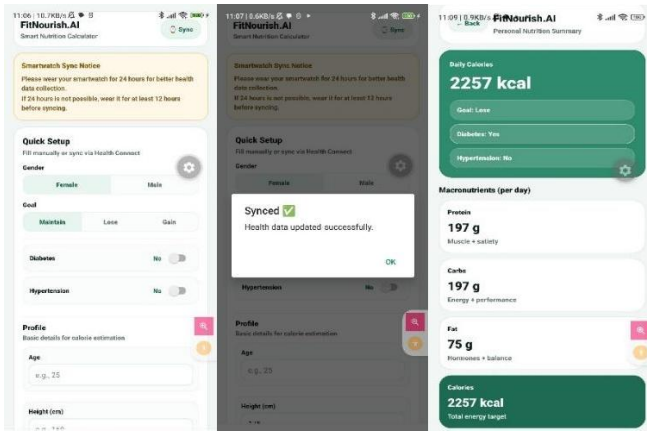


Fig.1. Nutrition calculation workflow showing user input, wearable data synchronization via Health Connect, and the final personalized nutrition output

### D. Predictive Health-Risk Classification

The health analytics module classifies users into **Low**, **Medium**, and **High**-risk categories. Candidate classifiers included Logistic Regression, Random Forest [15], Gradient Boosting, and XGBoost [16]. Input features included cholesterol level, blood pressure, glucose level, smoking status, exercise behavior, sleep duration, and selected activity-related indicators. XGBoost was selected as the final classifier due to its stronger validation performance and suitability for nonlinear structured data.

A 5-fold stratified cross-validation strategy was used to preserve class distribution during evaluation. Model performance was assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis. Feature importance analysis was also applied to identify the most influential health-risk predictors. The predicted risk level is used as decision support rather than as an isolated output, helping users interpret their nutrition targets more carefully based on their health condition.

### E. Grocery Product Matching

A Cross-retailer grocery comparison requires reliable product matching because supermarkets often describe similar products using different product names, brand formats, unit sizes, and categories. Therefore, product matching was formulated as an entity resolution task [10], [11]. For each candidate product pair, textual and structured similarity features were extracted, including TF-IDF cosine similarity, token similarity, brand similarity, category similarity, and unit-size compatibility.

Logistic Regression, Random Forest, and Multi-Layer Perceptron (MLP) models were evaluated for binary product matching. The model predicts whether two products from different retailers represent the same or sufficiently comparable item. The MLP model was selected when it achieved the strongest matching performance because it can capture nonlinear relationships among mixed textual and structured attributes. Only matched or comparable products were passed to the grocery optimization module to avoid unreliable price comparisons.

### F. Travel-Aware Grocery Optimization

The grocery optimization module recommends the most practical shopping strategy by considering product price, availability, and travel cost. For each grocery item required by the nutrition recommendation, the system retrieves the latest available price from each retailer. Stores that do not contain the required items are excluded from full-basket calculations.

Unlike simple price comparison systems, FitNourish.AI incorporates travel cost because the cheapest product is not always the best choice if the travel distance is high. Store distance was estimated using route-based distance where available, while the Haversine formula was used as a fallback approximation method [12]. The best single-store recommendation was selected by minimizing the combined basket and travel cost:

$$S^* = \arg \min_s (TotalCost_s)$$

Where  $S^*$  represents the optimal store,  $BasketCost_s$  the total item cost at store  $s$ , and  $TravelCost_s$  represents the estimated travel cost to reach that store.

The system also evaluates multi-store purchasing strategies inspired by the Traveling Purchaser Problem (TPP) [13]. A multi-store plan is accepted only when product savings are greater than the additional travel cost. If the additional travel cost exceeds the savings, the system recommends the best single-store option to avoid unnecessary travel. Since exhaustive evaluation of all product-store combinations is computationally expensive, a heuristic approach was used.

Candidate stores were filtered based on item availability and price competitiveness, and product assignment was performed using a greedy strategy that prioritizes lower item cost while considering travel impact.

#### IV. RESULTS AND DISCUSSION

This section evaluates the proposed **FitNourish.AI** framework across its four core modules: personalized nutrition prediction, health-risk classification, grocery product matching, and travel-aware grocery optimization. The results show that the system not only achieves strong predictive performance but also converts health and nutrition outputs into practical grocery purchasing decisions.

##### A. Nutrition Prediction Performance

The XGBoost-based multi-output regression model achieved strong performance in predicting personalized calorie and macronutrient requirements. Overall, the model recorded a **Mean Absolute Error (MAE) of 19.88**, **Root Mean Squared Error (RMSE) of 44.63**, and an  **$R^2$  score of 0.957**, indicating high explanatory power and low prediction error. A target-wise analysis further confirmed the consistency of the model.

TABLE II. TARGET-WISE PERFORMANCE OF THE NUTRITION PREDICTION MODEL

| Target Output | MAE        | RMSE       | ( $R^2$ ) |
|---------------|------------|------------|-----------|
| Calories      | 64.39 kcal | 88.13 kcal | 0.962     |
| Protein       | 3.84 g     | 5.51 g     | 0.954     |
| Carbohydrates | 8.98 g     | 12.65 g    | 0.949     |
| Fat           | 2.31 g     | 3.13 g     | 0.962     |

These results demonstrate that the model can effectively learn the relationship between user profile data, disease conditions, wearable activity features, and nutritional requirements. Compared with traditional formula-based estimation, the proposed model provides more adaptive predictions because it incorporates activity-related and health-related features rather than relying only on static demographic inputs.

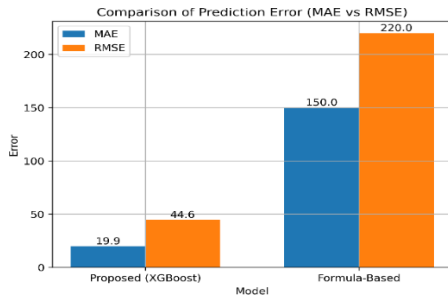


Fig.2.Performance of the XGBoost-based nutrition prediction model across calorie and macronutrient outputs.

##### B. Health-Risk Classification Performance

The health-risk classification module achieved a **test accuracy of 96.60%** and a **weighted F1-score of 0.97** in classifying users into Low, Medium, and High-risk categories. Among the evaluated models, XGBoost outperformed Logistic Regression, Random Forest, and Gradient Boosting, confirming its suitability for structured health-related tabular data.

The model showed particularly strong performance in identifying High-risk users, achieving **precision of 0.98** and **recall of 0.98**. This is important because misclassifying high-risk users may reduce the effectiveness of personalized health guidance. Analysis identified **cholesterol**, **blood pressure**, and **glucose level** as the most influential predictors, which aligns with common cardiometabolic risk indicators.

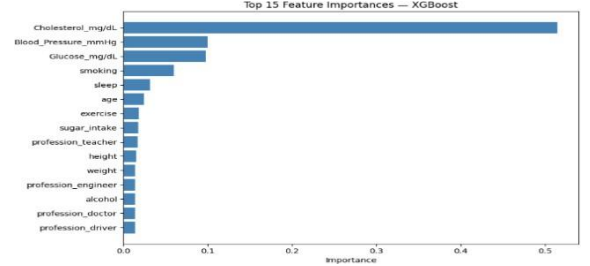


Fig.3. Health-risk classification results showing per- class performance and key importance

##### C. Product Matching Performance

The grocery product matching module was evaluated as a binary classification task to identify comparable products across different retailers. As shown in Fig. 4, the Multi-Layer Perceptron (MLP) achieved the best performance among the tested models, recording an accuracy of 0.875 and an F1-score of 0.779. This shows that the MLP model was more effective than Logistic Regression and Random Forest in handling product-matching features.

The MLP model also recorded a precision of 0.851 and a recall of 0.718. The higher precision indicates that the model reduced false matches, which is important because incorrect product matching can lead to misleading price comparisons and inaccurate grocery recommendations. These results show that nonlinear learning is useful for handling variations in product names, brands, unit sizes, and categories across Sri Lankan supermarket catalogues.

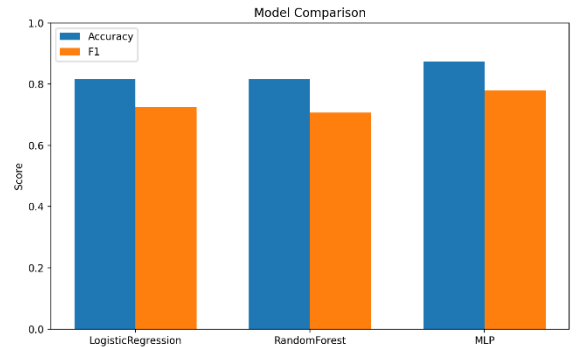


Fig.4.Product matching model comparison based on accuracy and F1-score.

##### D. Travel-Aware Grocery Optimization

The travel-aware grocery optimization module was evaluated using 30 shopping baskets by comparing single-store and multi-store purchasing strategies. Unlike basic price comparison, the proposed optimizer considers both product price and travel cost, since the cheapest product combination may not provide the lowest total cost when additional travel is included.

As shown in Fig. 5(a), the total cost difference between single-store and multi-store purchasing varies across baskets.



Multi-store purchasing is beneficial only when product-level savings are greater than the additional travel cost. Therefore, the system does not automatically select a multi-store plan; instead, it recommends the best single-store option when multi-store savings are insufficient. Fig. 5(b) shows that most optimized plans require only one or two store visits, indicating that the system avoids unnecessary shopping complexity while still identifying cost-saving opportunities.

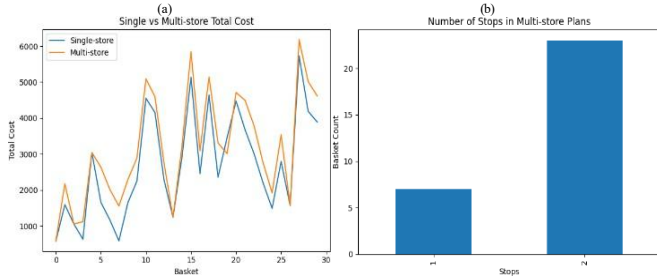


Fig.5. Travel-aware grocery optimization results: (a) total cost comparison between single-store and multi-store purchasing strategies, and (b) number of store visits in optimized plans.

#### E. Overall Discussion

The results demonstrate that FitNourish.AI provides practical personalized health support by linking nutrition prediction, health- risk classification, and grocery optimization in one framework. The system not only predicts calorie and macronutrient needs, but also supports realistic purchasing decisions using product availability, price, and travel cost.

Overall, FitNourish.AI addresses the gap between digital health recommendations and real-world food purchasing. Future work will focus on improving wearable data quality, expanding the dataset, adding more health indicators, and testing the system with a larger Sri Lankan user group.

### V. CONCLUSION AND FUTURE WORK

FitNourish.AI presents a unified AI-driven framework that integrates personalized nutrition prediction, health-risk classification, wearable-informed data, and travel-aware grocery optimization. The system addresses fragmentation in existing digital health solutions by linking calorie and macronutrient estimation with practical grocery purchasing decisions based on price, availability, and travel cost. The results demonstrate strong performance across nutrition prediction, risk stratification, product matching, and grocery optimization. Future work will expand the dataset, include additional disease conditions, retrain the models using larger and more diverse user profiles, improve real-time supermarket integration, and validate the system with a wider Sri Lankan user group to enhance robustness, scalability, and real-world applicability.

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